

Feature Engineering Guide

Football Top 5 Leagues Project

Purpose

Feature Engineering creates new variables from existing data to simplify the analysis and directly support our four research questions. These shared features will be reused during EDA, SQL analysis and the final presentation.

Recommended Workflow

Step	Description
Step 1	Create the Feature Engineering section and explain its purpose.
Step 2	Create match_result (Home Win / Draw / Away Win).
Step 3	Create goal_difference.
Step 4	Create total_goals.
Step 5	Create total_xg.
Step 6	Create xg_difference.
Step 7	Create points_difference.
Step 8	Create recent_form_difference.
Step 9	Create average_team_rating and rating_difference.
Step 10	Validate the new features.
Step 11	Document every feature in Markdown.

Feature Planning by Research Question

Feature	Q1	Q2	Q3	Q4	Purpose
match_result	✓	✓	✓	✓	Match outcome
goal_difference	✓		✓		Winning margin
total_goals	✓	✓		✓	Total goals
total_xg	✓			✓	Total Expected Goals
xg_difference	✓				Expected Goals comparison
points_difference			✓		Current points comparison
recent_form_difference		✓			Recent form comparison
average_team_rating				✓	Average team quality
rating_difference				✓	Compare team ratings

How Each Feature Is Built

Feature	Built From
match_result	home_score & away_score
goal_difference	home_score – away_score
total_goals	home_score + away_score
total_xg	home_xg + away_xg
xg_difference	home_xg – away_xg

points_difference	home_points – away_points
recent_form_difference	home_points_last_4_matches – away_points_last_4_matches
average_team_rating	Average of defense, midfield and attack ratings
rating_difference	Home average rating – Away average rating

Priority Order

Core Features: match_result, goal_difference, total_goals, total_xg, xg_difference.

Research-Specific Features: points_difference, recent_form_difference, average_team_rating and rating_difference.

Final Validation Checklist

- Validate the values of every engineered feature.
- Verify data types.
- Document every feature in Markdown.
- Ensure every team member uses the same engineered features.
- Export cleaned_data.csv only after the team agrees on the final cleaning decisions.